Rapid Communications

Anomalous Kerr effect in $SrRuO_3$ thin films

F. Michael Bartram,[1,2,*] Sopheak Sorn,[1,*] Zhuolu Li,[2,*] Kyle Hwangbo,[1] Shengchun Shen,[2] Felix Frontini,[1] Liqun He,[1] Pu Yu,[2,3,4,†] Arun Paramekanti,[1,‡] and Luyi Yang[1,2,3,§]

[1]*Department of Physics, University of Toronto, Toronto, Ontario, Canada M5S 1A7*

[2]*State Key Laboratory of Low Dimensional Quantum Physics, Department of Physics, Tsinghua University, Beijing 100084, China*

[3]*Frontier Science Center for Quantum Information, Beijing 100084, China*

[4]*RIKEN Center for Emergent Matter Science (CEMS), Wako 351-198, Japan*

(Received 23 August 2019; revised 9 October 2020; accepted 13 October 2020; published 26 October 2020)

We study the magneto-optical Kerr effect (MOKE) in $SrRuO_3$ thin films, uncovering wide regimes of wavelength, temperature, and magnetic field where the Kerr rotation is not simply proportional to the magnetization but instead displays two-component behavior. One component of the MOKE signal tracks the average magnetization, while the second "anomalous" component bears a resemblance to anomalies in the Hall resistivity which have been previously reported in skyrmion materials. We present a theory showing that the MOKE anomalies arise from the nonmonotonic relation between the Kerr angle and the magnetization, when we average over magnetic domains which proliferate near the coercive field. Our results suggest that inhomogeneous domain formation, rather than skyrmions, may provide a common origin for the observed MOKE and Hall resistivity anomalies.

DOI: 10.1103/PhysRevB.102.140408

Introduction. In magnetic solids, the nontrivial momentum space structure of Bloch bands [1–4] reveals itself via the intrinsic anomalous Hall effect which arises due to spin-orbit coupling and the band Berry curvature [5]. Remarkably, three-dimensional (3D) magnetic solids can host topological monopolelike singularities of the Berry curvature at Weyl nodes in the dispersion [6–9]. These Weyl nodes can contribute significantly to ρ_{xy} as seen in metallic antiferromagnets such as Mn_3X (X = Sn, Ge) [8,9], and in metallic ferromagnets such as $SrRuO_3$ where ρ_{xy} exhibits an unusual nonmonotonic temperature dependence as the Weyl nodes approach the Fermi energy via tuning magnetization or temperature [5,6,10].

Magnetic solids can also harbor real-space topological textures called skyrmions. Skyrmions imprint a real-space Berry curvature on conduction electrons, inducing a so-called "topological Hall effect" (THE) [11,12]. Skyrmion crystals and THE have been observed in various materials including MnSi [11,12], MnGe [13], Ir/Fe/Co/Pt multilayers [14–16], and Gd_2Pd_3Si [17]. In recent work, ultrathin epitaxial films of $SrRuO_3$ have been proposed as an oxide-based platform for hosting nanoscale skyrmions, stabilized by a strong chiral interfacial Dzyaloshinskii-Moriya (IDM) exchange [18,19].

Experimental support for skyrmions in ultrathin $SrRuO_3$ films comes from the observation of anomalous bumplike features in ρ_{xy} near the coercive field while traversing a magnetic hysteresis loop [18–23], similar to THE anomalies seen in other skyrmion materials. However, alternative proposals suggest that the Hall anomalies in this ultrathin limit can arise from the complicated temperature dependence of the intrinsic ρ_{xy}, together with atomic layer inhomogeneities in the ferromagnetic transition temperature T_c or the coercive field [24–27]. These alternative proposals rely on the momentum-space Berry curvature contribution to the intrinsic Hall effect, but the validity of such theories which simply add up the Hall response of two distinct regions remains unclear.

To assess the importance of momentum-space Berry curvature versus the role of skyrmions, and to further test the validity of these proposals, it is important to examine distinct regimes using new experimental probes. Here we study the magneto-optical Kerr effect (MOKE) in $SrRuO_3$ films. MOKE is a powerful, contactless, and nondestructive technique with high sensitivity and submicrometer spatial resolution. It has been widely used to study electronic and magnetic properties in magnetic materials and devices [28] and more recently in atomically thin 2D van der Waals magnets [29,30].

In contrast to previous work, our samples (30–200 nm thick) are far from the ultrathin limit, so the IDM interaction, skyrmions, and atomic layer inhomogeneities are not expected to play an important role. Remarkably, even in this regime, we discover bumplike anomalies in the MOKE signal over wide ranges of temperature, magnetic field, and frequencies, while the magnetization exhibits normal squarelike hysteresis loops. Significantly, this observation contradicts the well-established lore that the polar Kerr rotation is linearly proportional to the macroscopic magnetization in ferromagnetic thin films [18,21,28,31]. We describe a controlled theory for the high frequency response, showing that the MOKE anomalies can be semiquantitatively captured by a combination of the

*These authors contributed equally to this work.
†Corresponding author: yupu@mail.tsinghua.edu.cn
‡Corresponding author: arunp@physics.utoronto.ca
§Corresponding author: luyi-yang@mail.tsinghua.edu.cn

 ©2020 American Physical Society

nonmonotonic magnetization dependence of the Kerr angle and local averaging over magnetic domains. This experimental discovery of a Kerr anomaly, and its theoretical explanation, constitute the key new significant results of our work.

Experimental observations. $SrRuO_3$ films were grown using pulsed-laser deposition onto both $(LaAlO_3)_{0.3}(Sr_2TaAlO_6)_{0.7}$ (LSAT) and $SrTiO_3$ (STO) substrates. In contrast to previous studies which explored ultrathin few unit-cell films, our $SrRuO_3$ films range in thickness from 30 to 200 nm, displaying ferromagnetic order below $T_c \sim 150$ K [32]. Here, we focus on the case of an 88-nm-thick film grown on LSAT [see Supplemental Material (SM) [33] for data on other films]. We have measured the Kerr rotation in these films using a wavelength-tunable pulsed laser (repetition rate: 80 MHz) reduced to low power (<1 mW) and focused weakly onto an ~50 μm spot on the sample, with the reflected beam modulated by a photoelastic modulator and passed through a Wollaston prism into a pair of balanced photodiodes, allowing measurement of both the real and imaginary parts of the Kerr angle. Measurements were done in a typical polar Kerr configuration, with the beam reflected at normal incidence and the external field applied in the out-of-plane direction.

Figure 1 shows measurements of the Kerr rotation θ_K at normal incidence, Hall resistance R_{xy}, and magnetization M, at temperatures from 80 to 130 K for an 88-nm-thick $SrRuO_3$ sample on LSAT. We show and discuss only the real part of the Kerr rotation at $\lambda = 600$, 700, and 800 nm; the imaginary part, and additional data at 500 and 1500 nm, can be found in the SM [33]. While the magnetization (a) has the shape of a typical hysteresis loop, both R_{xy} (b), and the Kerr rotations at $\lambda = 600$ nm (c) and $\lambda = 700$ nm (d), show a large additional bumplike contribution similar to the topological contribution seen in systems hosting magnetic skyrmions. Such an additional contribution was also observed in previous R_{xy} measurements on few unit-cell ultrathin films, and was attributed to skyrmions [18–23]. For longer wavelengths such as $\lambda = 800$ nm (e) and $\lambda = 1500$ nm (see SM), however, this additional contribution is no longer present in our samples, and θ_K tracks the magnetization curve.

In addition to these anomalies, the values of R_{xy} and each θ_K at large fields, where the magnetization is uniform, change in very different ways as the temperature (and therefore the value of this uniform magnetization) is varied. To see this clearly, we show in (f) these values, which we define as $R_{xy,U}(T)$ and $\theta_{K,U}(T)$ plotted against the corresponding value of the saturation magnetization $M(T)$. (The subscript U denotes "uniform" since the high-field regime is expected to have a uniform magnetization across the sample.) For the data at $\lambda = 800$ nm and $\lambda = 1500$ nm, the result of this is approximately linear. Because at these wavelengths the hysteresis loops simply track the magnetization (with no anomalies), we conclude that $\theta_K \propto M$ across all temperatures and field values. For the R_{xy}, $\lambda = 600$ nm and $\lambda = 700$ nm data, however, there is a clearly nonlinear relationship, including a zero crossing at some $M > 0$ for each of them.

For the MOKE data in which anomalies are present, such as those in Figs. 1(c) and 1(d), the Kerr angle θ_K appears to exhibit two components: one component tracks the M

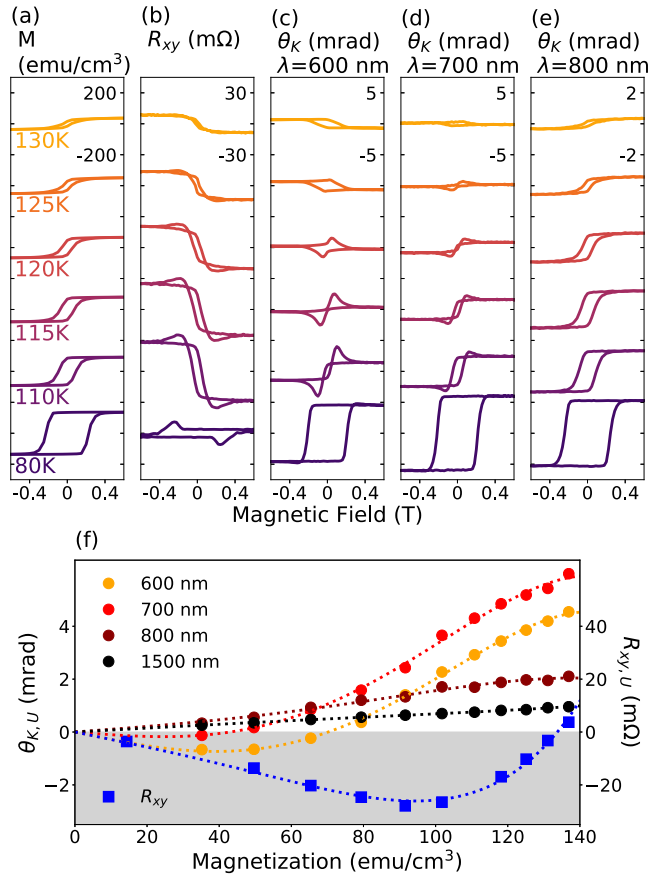

FIG. 1. Comparison of (a) magnetization M, (b) Hall resistance R_{xy}, and Kerr rotation θ_K at normal incidence with (c) 600, (d) 700, and (e) 800 nm lasers for an 88-nm-thick $SrRuO_3$ sample on LSAT. Both M and the 800 nm Kerr rotation data show a typical hysteresis loop. However, R_{xy} and θ_K at 600 and 700 nm show additional bump features. The bump features in θ_K are of opposite sign when compared to those in the R_{xy} measurements for the same field direction. Also shown in (f) are the values of R_{xy} and θ_K at high fields, where the magnetization is uniform, which we define as $R_{xy,U}(T)$ and $\theta_{K,U}(T)$, plotted against the corresponding saturation magnetization $M(T)$. Dotted lines are guides to the eye. Note that an additional data set at 1500 nm not shown above is present here; the hysteresis loops here are very similar to those at 800 nm (see SM [33] for details).

hysteresis loop (with the proportionality constant potentially changing with temperature, as discussed above), while the second component produces the bumplike anomaly. To see this more clearly, we can separate θ_K into a corresponding "normal contribution" $\theta_{K,M}$, and an additional contribution $\theta_{K,A}$, which we call the "anomalous Kerr angle," so that $\theta_K = \theta_{K,M} + \theta_{K,A}$. We illustrate this analysis in Fig. 2 for $\lambda = 600$ nm at $T = 105$ K, where we obtain $\theta_{K,M}$ by scaling directly measured magnetization data in order to cancel off the Kerr rotation at saturation magnetization [see dashed line in Fig. 2(a)]. Subtracting this from the $\lambda = 600$ nm Kerr signal loop (line with dots) gives a difference signal $\theta_K - \theta_{K,M}$ (blue shaded area) which defines the anomalous Kerr angle $\theta_{K,A}$. The full field and temperature dependence of $\theta_{K,A}$ quantified in this manner is displayed in Fig. 2(b); the dots indicate the coercive field B_c where the net magnetization vanishes.

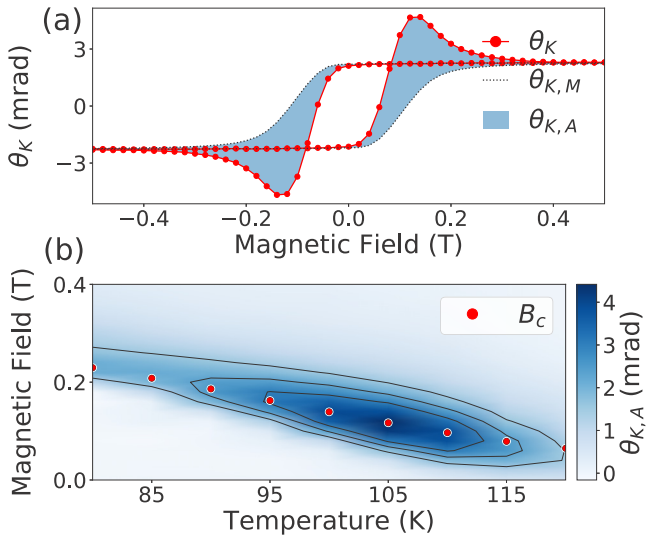

FIG. 2. (a) Kerr rotation measured with a 600 nm laser at $T = 105$ K for an 88 nm $SrRuO_3$ film on LSAT, showing the expected signal for a linear dependence on magnetization $\theta_{K,M}$, and the additional contribution $\theta_{K,A}$. (b) Color plot of the $\theta_{K,A}$ in the temperature-field plane. Red dots indicate B_C, the field strength where the net magnetization becomes zero.

Here we only show temperatures above 80 K; $\theta_{K,A}$ appears to persist down to very low temperatures but is difficult to clearly extract in this manner, as it becomes relatively small while $\theta_{K,M}$ becomes much larger. We observe that the largest $\theta_{K,A}$ occurs roughly around B_c. It is tempting to assign this feature to skyrmions as they form in the vicinity of magnetization reversals [19,34] in ultrathin films with chiral magnetic interactions. However, $SrRuO_3$ films in our thickness regime possess no known mechanism to stabilize skyrmions, making this an implausible explanation.

We clearly observe the frequency dependence of these anomalies with this analysis. Figure 3 shows both the Kerr rotation at uniform magnetization $\theta_{K,U}$ (as discussed earlier)

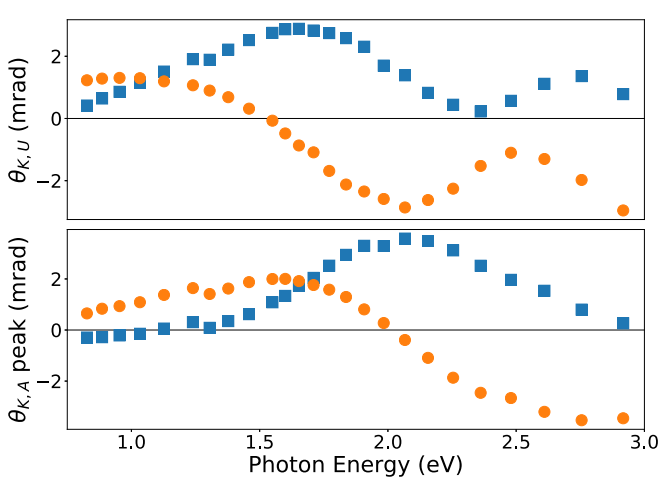

FIG. 3. Real (blue squares) and imaginary (orange circles) parts of the Kerr rotation at uniform magnetization $\theta_{K,U}$ (top) and the peak of the anomalous Kerr angle $\theta_{K,A}$ (bottom), shown as a function of frequency, for $T = 110$ K.

and the peak value of the anomalous component, $\theta_{K,A\,\mathrm{peak}}$, as a function of the photon energy of the laser. Here we also show the imaginary part of the Kerr angle (see SM for more details). We note that the appearance of the anomalies exhibits a resonant behavior, with a peak at around $\lambda = 600$ nm (2 eV) in the real part, and a concomitant zero crossing in the imaginary part consistent with Kramers-Kronig relations.

Similar measurements on films of varying thicknesses yield a nonzero $\theta_{K,A}$ for thicknesses in the range ~70–120 nm (see SM [33] for additional data). In some thinner films (~30 nm), we observe anomalies in R_{xy}, but not in any of our MOKE measurements. However, in these cases we also do not observe any zero crossings of $\theta_{K,U}$ at nonzero magnetization for the measured λ, as for the long-wavelength data in the 88 nm film. (It is possible that the anomalies may appear at λ outside the range accessed in our study.) Finally, films grown on LSAT which has greater lattice mismatch (1.4%) with $SrRuO_3$ when compared with the STO substrate (lattice mismatch: 0.45%), exhibit a larger, and broader (in field), $\theta_{K,A}$. Thus strain has a strong impact on the observed anomalies.

In previous work on R_{xy} anomalies in ultrathin $SrRuO_3$ films, it was suggested that an alternative source (other than skyrmions) for the bumps could be the temperature-dependent sign changes in the bulk anomalous Hall effect combined with inhomogeneities in T_c or the coercive field across the sample [24]. While such inhomogeneities could be important in few unit-cell ultrathin films, where they were proposed to originate from single unit-cell variations in the film thickness, it is less clear that such atomic scale variations can impact the relatively thick films studied in our work. However, a ubiquitous feature common to such magnetic thin films is magnetic domain proliferation near the coercive field [19,35]. We thus turn to a phenomenological theory of the MOKE in the presence of such domains.

Theory. As we pass through the coercive field B_c during a magnetization reversal process, minority magnetic domains start to proliferate, and eventually take over the system. During such a process, let $\mathcal{M}(B, \mathbf{r})$ be the inhomogeneous local magnetization (perpendicular to the film) at a point $\mathbf{r}$ in a field B. The average magnetization $M(B) = (1/V)\int d^3\mathbf{r}\,\mathcal{M}(B, \mathbf{r})$, where V denotes the sample volume. Let the Kerr angle in a system with *uniform* magnetization M be given by $\theta_{K,U}(M)$. We then propose that the measured Kerr angle $\theta_K(B) = (1/V)\int d^3\mathbf{r}\,\theta_{K,U}[\mathcal{M}(B, \mathbf{r})]$; this averaging results in a nontrivial behavior when $\theta_{K,U}(M)$ is a nonlinear function. A heuristic justification for this local magnetization approximation (LMA) for θ_K is that the electronic response at high frequency ω must be local in space. Semiclassically, electrons with Fermi velocity v_F traverse only a distance $\ell_\omega \sim \pi v_F/\omega$ in a half-period π/ω; using $v_F \sim 2 \times 10^5$ m/s [36] and $\hbar\omega \sim 2$ eV, yields $\ell_\omega \sim 2$ Å, i.e., on the scale of the lattice spacing. In the SM [33], we compare the LMA with the numerically computed frequency-dependent conductivity tensor from the Kubo formula for a t_{2g} model with spin-orbit coupling using various inhomogeneous magnetization profiles, showing that they quantitatively agree for $\hbar\omega \gtrsim W/4$, where the t_{2g} bandwidth $W \sim 2$–3 eV [6].

We can then compute the Kerr loop in three steps. (i) We simplify the inhomogeneous magnetization profile

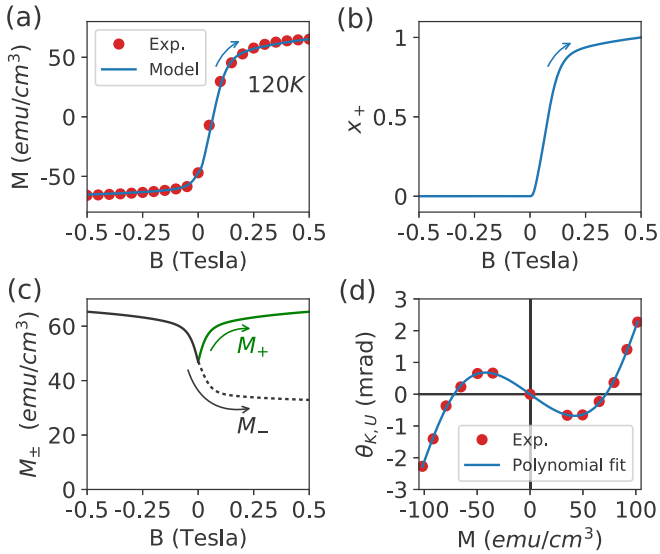

FIG. 4. (a) Magnetization data (dots) at $T = 120$ K in an increasing-field sweep for 88 nm SRO film. Solid line is a fit to the two-domain model $M = x_+M_+ - x_-M_-$ discussed in the text and SM [33]. Extracted domain fraction x_+, and domain magnetizations $M_\pm$ are shown in (b) and (c), with arrows indicating the direction of the field sweep. (d) Kerr angle $\theta_{K,U}$ for $\lambda = 600$ nm as a function of uniform magnetization for 88 nm film; dots are data points obtained in a large applied field at different temperatures; solid line shows a fit.

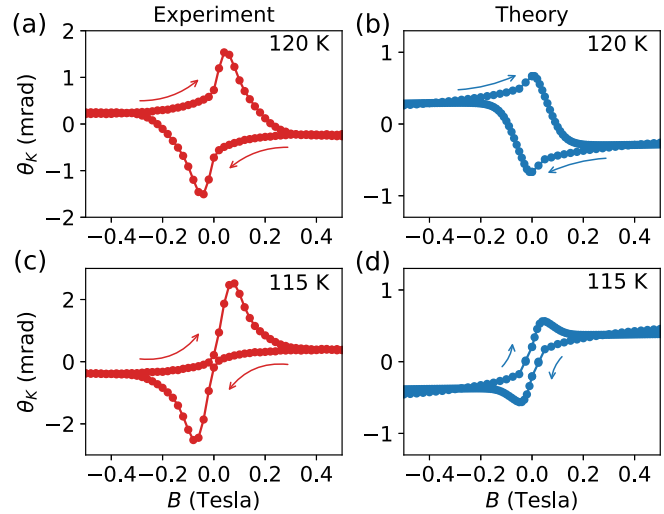

FIG. 5. (a) and (c) depict Kerr anomalies seen in experiments on the 88 nm $SrRuO_3$ sample using a 600 nm laser at 120 and 115 K, respectively. Arrows denote the direction of the field sweep. (b) and (d) show the theoretically computed Kerr anomalies within the LMA, displaying bump features which agree qualitatively with the experiment. The anomalies arise from magnetic domains and the nonmonotonic magnetization dependence of $\theta_{K,U}$ fits shown in Fig. 4(d).

$\mathcal{M}(B, \mathbf{r})$ by two domain types, with magnetizations $M_+(B)$ and $-M_-(B)$ perpendicular to the film, and corresponding volume fractions $x_+(B)$ and $x_-(B) = 1 - x_+(B)$. Here $M_\pm \geqslant 0$ and $0 \leqslant x_\pm \leqslant 1$. This yields $M(B) = x_+(B)M_+(B) - x_-(B)M_-(B)$. We fit the magnetization data to extract x_+, M_+, M_- as a function of B. (ii) Next, we assume that the Kerr angle at saturation magnetization is a good proxy for $\theta_{K,U}$. We thus fit the saturation Kerr angle as a function of saturation magnetization to get $\theta_{K,U}(M)$. (iii) Finally, using these fitted functions as inputs to the LMA, we obtain the theoretical estimate for the average Kerr angle as $\theta_K(B) = x_+(B)\theta_{K,U}[M_+(B)] + x_-(B)\theta_{K,U}[-M_-(B)]$.

Figure 4(a) shows an example of a fit to the magnetization data in an increasing-field sweep (see SM [33] for details). We make a simple ansatz for x_+, with $x_+(B < 0) = 0$, and $x_+ \to 1$ for large $B > 0$. Thus, $M_-(B < 0) = -M$ which can be read off from the measured M, shown as the solid black line in Fig. 4(c). Symmetry dictates $M_+(B > 0) = M_-(B < 0)$, as shown by the solid green line. These two constraints, along with a reasonable ansatz for $M_-(B > 0)$, allow us to fit $M(B > 0)$, and extract $x_+(B > 0)$ and $M_-(B > 0)$ as shown in Figs. 4(b) and 4(c). Our results below are robust against variations in the precise shape of $M_-(B > 0)$.

We next turn to the saturation Kerr angle, $\theta_{K,U}(M)$, which is measured at a large enough field $|B| \sim 0.5$ T to achieve saturation at a given temperature, and then varying the temperature to tune the saturation value of M. Figure 4(d) shows a polynomial fit to $\theta_{K,U}(M)$, which exhibits a nonmonotonic behavior similar to the Weyl-node induced nonmonotonic Hall resistivity [5,6,10].

These key results from the fits in Fig. 4 serve as inputs to the LMA. Figure 5 compares the experimentally measured θ_K and the LMA theory results over the hysteresis loop, showing good qualitative agreement in the overall shape and magnitude of the observed bumps. Achieving a better quantitative agreement in terms of the height of the bumps and the precise shapes requires a theory for $\theta_{K,U}(M)$ at each temperature, while we have extracted it using experimental values from different temperature datasets (see SM [33] for further discussion of this point).

Such an effective medium approximation fails to explain the anomalies we observe in R_{xy} [33]; this discrepancy may be due to the fact that the dc conductivity tensor is nonlocal and expected to be more sensitive to details of the domain size distribution and domain walls. Using a t_{2g} model Hamiltonian, we show in the SM [33] that domain walls introduce a correction to ρ_{xy} which can potentially account for the Hall anomalies.

Conclusion. We have measured an anomalous contribution to the Kerr signal in thin films of $SrRuO_3$ over wide regimes of magnetic field, temperature, and laser wavelength, which appears to behave similarly to R_{xy} anomalies attributed to the THE in skyrmion materials. We have instead shown that these anomalies in the Kerr signal can arise from the nonlinear dependence of the Kerr angle on magnetization, together with magnetic domain formation during magnetization reversal. For bulk Hall transport, previous work has shown that the nonmonotonic behavior of the Hall resistivity as a function of magnetization or temperature can arise from Weyl nodes in the band structure [5,10]. Weyl nodes, and their interplay with magnetic domain walls, might thus provide a microscopic origin for the nonlinear Kerr signal as well as the observed dc Hall anomalies. Such an interplay has been recently examined

for R_{xy} in the antiferromagnetic Weyl metals Mn_3Ge and Mn_3Sn [37,38]. In light of our results, it would be valuable to also reevaluate the role of magnetic domains versus skyrmions in ultrathin $SrRuO_3$ films.

Acknowledgments. The optical measurements were performed at Tsinghua University and at the University of Toronto and were supported by the Tsinghua University Startup Fund, the CIFAR Azrieli Global Scholars Programme, NSERC Canada Research Chair, the Canadian Foundation for Innovation, and the Ontario Research Fund. The theoretical studies were performed at the University of Toronto and were funded by NSERC of Canada. This research was enabled in part by support provided by WestGrid [45] and Compute Canada Calcul Canada [46]. Sample growth and characterization were carried out at Tsinghua University and were supported by the Basic Science Center Project of NSFC under Grants No. 51788104 and No. 51872155; the National Basic Research Program of China under Grant No. 2016YFA0301004; and the Beijing Advanced Innovation Center for Future Chip (ICFC).

[1] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, Phys. Rev. Lett. **49**, 405 (1982).
[2] A. P. Schnyder, S. Ryu, A. Furusaki, and A. W. W. Ludwig, Phys. Rev. B **78**, 195125 (2008).
[3] A. Kitaev, *Periodic Table for Topological Insulators and Superconductors*, *AIP Conf. Proc. No.* 1134 (AIP, Melville, NY, 2009), p. 22.
[4] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, New J. Phys. **12**, 065010 (2010).
[5] N. Nagaosa, J. Sinova, S. Onoda, A. H. MacDonald, and N. P. Ong, Rev. Mod. Phys. **82**, 1539 (2010).
[6] Z. Fang, N. Nagaosa, K. S. Takahashi, A. Asamitsu, R. Mathieu, T. Ogasawara, H. Yamada, M. Kawasaki, Y. Tokura, and K. Terakura, Science **302**, 92 (2003).
[7] X. Wan, A. M. Turner, A. Vishwanath, and S. Y. Savrasov, Phys. Rev. B **83**, 205101 (2011).
[8] H. Yang, Y. Sun, Y. Zhang, W.-J. Shi, S. S. P. Parkin, and B. Yan, New J. Phys. **19**, 015008 (2017).
[9] J. Kübler and C. Felser, Europhys. Lett. **108**, 67001 (2014).
[10] Y. Chen, D. L. Bergman, and A. A. Burkov, Phys. Rev. B **88**, 125110 (2013).
[11] A. Neubauer, C. Pfleiderer, B. Binz, A. Rosch, R. Ritz, P. G. Niklowitz, and P. Böni, Phys. Rev. Lett. **102**, 186602 (2009).
[12] M. Lee, W. Kang, Y. Onose, Y. Tokura, and N. P. Ong, Phys. Rev. Lett. **102**, 186601 (2009).
[13] N. Kanazawa, Y. Onose, T. Arima, D. Okuyama, K. Ohoyama, S. Wakimoto, K. Kakurai, S. Ishiwata, and Y. Tokura, Phys. Rev. Lett. **106**, 156603 (2011).
[14] A. Soumyanarayanan, M. Raju, A. L. Gonzalez Oyarce, A. K. C. Tan, M.-Y. Im, A. P. Petrović, P. Ho, K. H. Khoo, M. Tran, C. K. Gan, F. Ernult, and C. Panagopoulos, Nat. Mater. **16**, 898 (2017).
[15] W. Legrand, J.-Y. Chauleau, D. Maccariello, N. Reyren, S. Collin, K. Bouzehouane, N. Jaouen, V. Cros, and A. Fert, Sci. Adv. **4**, eaat0415 (2018).
[16] J. A. Garlow, S. D. Pollard, M. Beleggia, T. Dutta, H. Yang, and Y. Zhu, Phys. Rev. Lett. **122**, 237201 (2019).
[17] T. Kurumaji, T. Nakajima, M. Hirschberger, A. Kikkawa, Y. Yamasaki, H. Sagayama, H. Nakao, Y. Taguchi, T.-h. Arima, and Y. Tokura, Science **365**, 914 (2019).
[18] J. Matsuno, N. Ogawa, K. Yasuda, F. Kagawa, W. Koshibae, N. Nagaosa, Y. Tokura, and M. Kawasaki, Sci. Adv. **2**, e1600304 (2016).
[19] K.-Y. Meng, A. S. Ahmed, M. Baćani, A.-O. Mandru, X. Zhao, N. Bagués, B. D. Esser, J. Flores, D. W. McComb, H. J. Hug, and F. Yang, Nano Lett. **19**, 3169 (2019).
[20] B. Pang, L. Zhang, Y. B. Chen, J. Zhou, S. Yao, S. Zhang, and Y. Chen, ACS Appl. Mater. Interfaces **9**, 3201 (2017).
[21] Y. Ohuchi, J. Matsuno, N. Ogawa, Y. Kozuka, M. Uchida, Y. Tokura, and M. Kawasaki, Nat. Commun. **9**, 213 (2018).
[22] L. Wang, Q. Feng, Y. Kim, R. Kim, K. H. Lee, S. D. Pollard, Y. J. Shin, H. Zhou, W. Peng, D. Lee, W. Meng, H. Yang, J. H. Han, M. Kim, Q. Lu, and T. W. Noh, Nat. Mater. **17**, 1087 (2018).
[23] Q. Qin, L. Liu, W. Lin, X. Shu, Q. Xie, Z. Lim, C. Li, S. He, G. M. Chow, and J. Chen, Adv. Mater. **31**, 1807008 (2019).
[24] D. Kan, T. Moriyama, K. Kobayashi, and Y. Shimakawa, Phys. Rev. B **98**, 180408(R) (2018).
[25] A. Gerber, Phys. Rev. B **98**, 214440 (2018).
[26] L. Wang, Q. Feng, H. G. Lee, E. K. Ko, Q. Lu, and T. W. Noh, Nano Lett. **20**, 2468 (2020).
[27] G. Malsch, D. Ivaneyko, P. Milde, L. Wysocki, L. Yang, P. H. M. van Loosdrecht, I. Lindfors-Vrejoiu, and L. M. Eng, ACS Appl. Nano Mater. **3**, 1182 (2020).
[28] Z. Q. Qiu and S. D. Bader, Rev. Sci. Instrum. **71**, 1243 (2000).
[29] B. Huang, G. Clark, E. Navarro-Moratalla, D. R. Klein, R. Cheng, K. L. Seyler, D. Zhong, E. Schmidgall, M. A. McGuire, D. H. Cobden, W. Yao, D. Xiao, P. Jarillo-Herrero, and X. Xu, Nature (London) **546**, 270 (2017).
[30] C. Gong, L. Li, Z. Li, H. Ji, A. Stern, Y. Xia, T. Cao, W. Bao, C. Wang, Y. Wang, Z. Q. Qiu, R. J. Cava, S. G. Louie, J. Xia, and X. Zhang, Nature (London) **546**, 265 (2017).
[31] P. N. Argyres, Phys. Rev. **97**, 334 (1955).
[32] Z. Li, S. Shen, Z. Tian, K. Hwangbo, M. Wang, Y. Wang, F. M. Bartram, L. He, Y. Lyu, Y. Dong, G. Wan, H. Li, N. Lu, J. Zang, H. Zhou, E. Arenholz, Q. He, L. Yang, W. Luo, and P. Yu, Nat. Commun. **11**, 184 (2020).
[33] See Supplemental Material at http://link.aps.org/supplemental/10.1103/PhysRevB.102.140408 for details of (i) additional MOKE data for the 88 nm sample at different laser frequencies, (ii) further MOKE data on different thickness films on STO and LSAT substrates, (iii) MOKE measurements at oblique incidence, (iv) discussion of why an alternative "nonintrinsic" explanation of the data in terms of inhomogeneities does not work, (v) numerical check of the validity of LMA at large frequencies, (vi) ansatz for the two-domain model, (vii) discussion on how a possible isothermal $\theta_{K,U}$-M curve may result in a quantitatively better agreement between the experiment and the theory, and (viii) dc Hall conductivities of a t_{2g} cubic

model with magnetic domain walls, showing a domain-wall correction to the Hall conductivity. Includes additional references [6,10,24,31,37–44] .

[34] M. Raju, A. Yagil, A. Soumyanarayanan, A. K. C. Tan, A. Almoalem, F. Ma, O. M. Auslaender, and C. Panagopoulos, Nat. Commun. **10**, 696 (2019).

[35] M. Zahradník, K. Uhlířová, T. Maroutian, G. Kurij, G. Agnus, M. Veis, and P. Lecoeur, Mater. Des. **187**, 108390 (2020).

[36] D. J. Singh, J. Appl. Phys. **79**, 4818 (1996).

[37] J. Liu and L. Balents, Phys. Rev. Lett. **119**, 087202 (2017).

[38] X. Li, C. Collignon, L. Xu, H. Zuo, A. Cavanna, U. Gennser, D. Mailly, B. Fauqué, L. Balents, Z. Zhu, and K. Behnia, Nat. Commun. **10**, 3021 (2019).

[39] G. Koster, L. Klein, W. Siemons, G. Rijnders, J. S. Dodge, C.-B. Eom, D. H. A. Blank, and M. R. Beasley, Rev. Mod. Phys. **84**, 253 (2012).

[40] G. Mahan, *Many-Particle Physics*, Physics of Solids and Liquids (Springer Publishing, New York, 2000); P. Coleman, *Introduction to Many-Body Physics* (Cambridge University Press, Cambridge, UK, 2015); D. J. Scalapino, S. R. White, and S. Zhang, Phys. Rev. B **47**, 7995 (1993).

[41] P. Kostic, Y. Okada, N. C. Collins, Z. Schlesinger, J. W. Reiner, L. Klein, A. Kapitulnik, T. H. Geballe, and M. R. Beasley, Phys. Rev. Lett. **81**, 2498 (1998).

[42] J. S. Dodge, C. P. Weber, J. Corson, J. Orenstein, Z. Schlesinger, J. W. Reiner, and M. R. Beasley, Phys. Rev. Lett. **85**, 4932 (2000).

[43] D. Stroud, Phys. Rev. B **12**, 3368 (1975).

[44] A. Granovsky, A. Vedyayev, and F. Brouers, J. Magn. Magn. Mater. **136**, 229 (1994).

[45] www.westgrid.ca.

[46] www.computecanada.ca.